

Problem Set

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Name: _____

1. (20 points) **Principal-Agent Problem with Moral Hazard**

A principal hires an agent to complete a project. The agent exerts effort e . Effort costs the agent $\frac{e^2}{2}$. Both, principal and agent, are risk neutral.

The project generates revenue $x = 10e$ for the principal. The principal offers a contract $w = a + bx$ where a is the base salary and $b \leq 1$ is the bonus rate.

The agent's reservation utility is $\underline{u} = 2$.

- (10 points) If effort is observable (first-best), what effort level maximizes total surplus? What contract would implement this?
- (10 points) If effort is not observable (second-best), *i.e.* $x = 10e + u$ with $u \stackrel{iid}{\sim} (0, \sigma_u^2)$, what effort will the agent choose if offered $b = 0.2$? Is this optimal for the principal?

2. (20 points) **Bayesian Games and Incomplete Information**

Two countries are negotiating a trade agreement. Country A knows its type (high-value $\theta_H = 10$ or low-value $\theta_L = 4$), but Country B only knows that each type is equally likely ($p = 0.5$).

Country A can propose either a comprehensive agreement (C) or a limited agreement (L). Country B then accepts (A) or rejects (R).

Payoffs:

- If type θ_H proposes C and B accepts: (8, 6)
 - If type θ_H proposes C and B rejects: (0, 0)
 - If type θ_H proposes L and B accepts: (5, 4)
 - If type θ_H proposes L and B rejects: (1, 0)
 - If type θ_L proposes C and B accepts: (3, 6)
 - If type θ_L proposes C and B rejects: (0, 0)
 - If type θ_L proposes L and B accepts: (4, 4)
 - If type θ_L proposes L and B rejects: (1, 0)
- (10 points) Is there a separating Perfect Bayesian Equilibrium where type θ_H plays C and type θ_L plays L? What would B's best response be?
 - (10 points) Is there a pooling Perfect Bayesian Equilibrium where both types play C? What would B's best response be?

3. (30 points) Consider the Dynamic Inconsistency of the Fiscal Policy game. Devise and write down the **extensive form** of the game. Think on discrete space (high tax τ^h , low tax τ^l , high investment I_h , low investment I_l) etc. You are free to choose how many stages the game has, the payoffs, etc. The idea of this question is to see if you can think on a situation/phenomena and model it using what we covered in this course.

4. (30 points) Suppose the government is planning on giving a subsidy for poor people. The utility function of every citizen is $u(c) = \sqrt{c}$. Suppose there are two kinds of people, poor people and rich people.
- (a) (10 points) If the only thing necessary to claim the subsidy was to tell the government you are poor, what is going to happen?
 - (b) (20 points) What can do the government to separate both types of people? Suppose rich people has wealth equal to $w_h = 100$, and poor people has wealth $w_l = 0$. Suppose the opportunity cost of doing paperwork is $c(p|w) = w * p^2$, *i.e.*, the opportunity cost is increasing in the level of wealth. What is the amount of necessary paperwork to claim the subsidy p to prevent rich people from claiming the subsidy? Will poor people claim the subsidy?